

Physics Olympiad: Computational Challenge

**2024:
Extensions**

BpHO: Extension challenges

There are 4 extension challenges.

Only two of which are physical models, being the varying air resistance and the planetary model.

The other extensions are the Graphical User Interface and the derivation paper.

EX TASK 1

Extension task 1 is a continuation of task 9, the resistive Verlet method. The extension, however, is that air resistance varies with altitude.

In the real world, air resistance varies with altitude as air density increases exponentially, the closer the air particle 'layer' is to the surface of earth.

Task 9 Verlet Method

Iteration n :

$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} -\frac{K v_x^2}{2m} \\ -gm - \frac{K v_y^2}{2m} \end{pmatrix}$$

$$v_{n+1} = v_n + a \Delta t$$

And plot the position x and position y to create our displacement graph.

Modifications

The value of K is determined by:

$$K = \rho C A$$

Where ρ is the fluid density. To apply a varying air resistance, we may recalculate K with a different ρ value, proportional to the altitude of the particle.

Fluid Density Formula

According to [this model](#), the fluid density of air, changes according to:

Where h is the altitude of the projectile.

New Verlet Method

$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} -\frac{K v_x^2}{2m} \\ -gm - \frac{K v_y^2}{2m} \end{pmatrix}$$

$$v_{n+1} = v_n + a \Delta t$$

Where $p_{n+1}^{(y)}$ is the y value of the position of the projectile, therefore the altitude.

The model varies **only** with altitude, and not any other variables such as temperature.

EX TASK 2

Extension task 2 is an incredibly complex problem. To get it to its current state, I have to majorly abstract certain pieces of the model, or I did not have enough time to implement them.

Model Restrictions

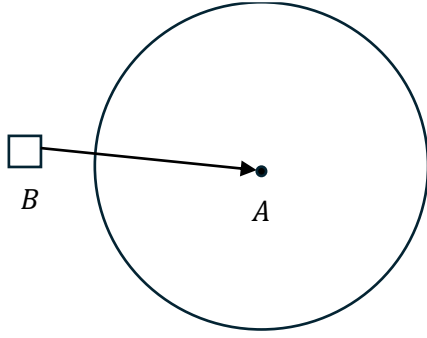
- Gravity does not change with distance to the planet
- No Coriolis effect (Earth rotation force)
- The particle does not experience air resistance when it otherwise would.

The axes of gravitational pull upon a moving particle

Since the particle is moving at incredibly fast speeds and high altitudes, the axis upon which the weight of the particle acts varies.

The force of attraction should always be between the centre of the masses. Since we have a particle, which only has one point of mass, and a planet where the centre may assumed to be the origin, we need to find a directional vector between these two masses.

Deriving the gravity vector



Despite the diagram, the model is **3D**

The vector $\overrightarrow{AB}$ is the vector upon which the gravity will be applied. This is relatively simple.

$$g \frac{[B - A]}{|\overrightarrow{AB}|} = a$$

Where a is the acceleration vector. We find the directional vector between A and B , normalise, and then multiply it by the gravitational constant.

Planetary Verlet Method

Since acceleration is constantly changing (the magnitude is the same, but the direction is different), a Verlet method seems like the most suitable modelling method, except, of course, we need to add a third dimension.

$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$v_{n+1} = v_n + a \Delta t$$

Above is the standard Verlet method.

$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$v_{n+1} = v_n + a \Delta t$$

$$g \frac{[B - A]}{|\overrightarrow{AB}|} = a$$

However, since point A is always at the origin, and B is the vector of the current position, we can model it as:

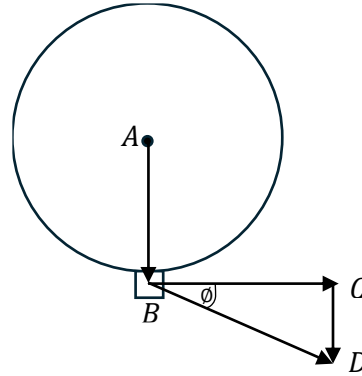
$$p_{n+1} = p_n + v_n \Delta t + \frac{1}{2} a_n (\Delta t)^2$$

$$a = g \frac{p_{n+1}}{|p|}$$

$$v_{n+1} = v_n + a \Delta t$$

Modelling velocity from launch

Launch velocity must be calculated slightly differently from usual, due to the fact we may be starting from any point on the surface of the planet, and at an angle.



So, we must find our velocity vector V . v is the magnitude of velocity.

A is known. B is known.

Let C be a vector perpendicular to $\overrightarrow{AB}$, but of the same magnitude.

$$\alpha \overrightarrow{AB} = \overrightarrow{CD}$$

$$\tan(\phi) |\overrightarrow{BC}| = |\overrightarrow{CD}|$$

$$\frac{|\overrightarrow{CD}|}{|\overrightarrow{AB}|} = \alpha$$

$$\overrightarrow{BC} + \alpha \overrightarrow{AB} = \overrightarrow{BD}$$

$$\frac{\overrightarrow{BD}}{|\overrightarrow{BD}|} v = V$$

Unfortunately, due to time constraints, I was unable to further add to the model. Besides the addition of latitude and longitude conversion, nothing more was added.